

2013 Notes from the Marking Centre – Chemistry

Introduction

This document has been produced for the teachers and candidates of the Stage 6 Chemistry course. It contains comments on candidate responses to the 2013 Higher School Certificate examination, highlighting their strengths in particular parts of the examination and indicating where candidates need to improve.

This document should be read along with:

- the [Chemistry Stage 6 Syllabus](#)
- the [2013 Higher School Certificate Chemistry examination \(PDF\)](#)
- the [marking guidelines \(PDF\)](#)
- [advice for candidates attempting science examinations](#)
- [Advice for HSC students about examinations](#)
- other support documents developed by the Board of Studies, Teaching and Educational Standards NSW to assist in the teaching and learning of Chemistry in Stage 6.

Section I – Part B

Candidates showed strength in these areas:

- balancing equations and using formulae (Qs.22, 23, 25)
- calculating moles and mass (Q.23)
- knowledge of the use of acids and indicators (Qs.21, 25)
- extracting data from graphs, tables and flow charts (Qs.22, 27, 30)
- describing and applying Le Chatelier's Principle (Q.25)
- comparing biopolymers and fossil-fuel-based polymers (Q.31)
- understanding that a positive E^0 value indicates the spontaneity of a reaction (Q.29).

Candidates need to improve in these areas:

- drawing and labelling diagrams (Q.21b)
- identifying ions and writing net ionic equations (Q.22)
- calculating E^0 for reactions, especially where non-metals are involved (Q.29)
- describing correct titration technique
- using specific language, for example 'this resulted in an increase/decrease in the HCl concentration', rather than 'this affected the HCl concentration' (Q.28).

Section II – Options

Question 32 – Industrial Chemistry

Candidates showed strength in these areas:

- identifying products of electrolysis and hazards presented by those products (part d)
- equilibrium considerations of the contact process (part e)
- constructing an I.C.E. table to calculate equilibrium constant (part b)
- describing the action of soaps and detergents (part a).

Candidates need to improve in these areas:

- including only relevant information about chemical processes
- understanding that maximising the yield reduces environmental impact in contact process (part e).

Question 33 – Shipwrecks, Corrosion and Conservation

Candidates showed strength in these areas:

- knowledge of corrosion protection
- knowledge of the work of scientists in this area (part e).

Candidates need to improve in these areas:

- understanding the difference between the 'solubility of oxygen' and the 'concentration of oxygen'.

Question 34 – The Biochemistry of Movement

Candidates showed strength in these areas:

- describing the structure and function of ATP (part a)
- explaining the effect of temperature on enzymes (part d).

Candidates need to improve in these areas:

- describing the structure of a peptide bond (part b)
- identifying the amine group in amino acids (part b)
- calculating energy (part c).

Question 35 – The Chemistry of Art

Candidates showed strength in these areas:

- understanding of emission spectra (part a)
- understanding of oxidation states (part c).

Candidates need to improve in these areas:

- understanding of bonding in complex ions and electron dot diagrams (part c)
- using specific terminology such as orbitals, subshells, and types of spectra.

Question 36 – Forensic Chemistry

Candidates showed strength in these areas:

- describing a test for an organic compound (part d)
- analysing chromatograms (part c).

Candidates need to improve in these areas:

- describing a laboratory procedure to produce and analyse an emission spectrum (part c)
- describing both chromatography and electrophoresis and making a justification based on their evidence (part e).